

Document Control

- System Name: Application
- Version: 1.0
- Author: Data Engineering Team
- Date: 2024-06-12
- Status: Draft

Executive Summary

- The Application enables automated ETL pipeline management, processing source CSV data to Delta tables through Delta Live Tables, with robust transformations, cleaning, aggregation, and SCD Type 2 historical data handling.

Business Objectives

- Automate data ingestion from CSV files to Delta tables
- Enforce consistent schema for customer and order data
- Ensure data quality via removal of nulls and duplicates
- Aggregate and track customer spend with historical auditing
- Implement reliable change data capture with SCD Type 2

In-Scope

- Source CSV data read and delta table load
- Schema enforcement for customer and order tables
- Data transformation and cleaning
- Ordersummary and customeraggregatespend table creation
- Aggregation logic on order data
- SCD Type 2 implementation for ordersummary
- Workflow orchestration using Delta Live Tables

Out-of-Scope

- Manual data manipulation
- Integration with external systems beyond defined delta tables
- Real-time or streaming ingestion mechanisms
- Reporting dashboards or visualization

Stakeholders

- Data Engineering Team
- Business Analysts

- Compliance Team
- Data Governance

Business Requirements

BR-ETL-001 — Load Source Customer Data

- Description: Read customer CSV data and load to delta table
/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- Business Value: Automated, structured customer data ingestion
- Priority: High

BR-ETL-002 — Load Source Order Data

- Description: Read order CSV data and load to delta table
/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Business Value: Automated, structured order data ingestion
- Priority: High

BR-ETL-003 — Customer Table Schema Enforcement

- Description: Enforce schema with fields CustId, Name, EmailId, Region for customer table
- Business Value: Consistency and data quality assurance
- Priority: High

BR-ETL-004 — Order Table Schema Enforcement

- Description: Enforce schema with fields OrderId, ItemName, PricePerUnit, Qty, Date, CustId for order table
- Business Value: Consistency and data quality assurance
- Priority: High

BR-ETL-005 — Calculate TotalAmount Field

- Description: Add column TotalAmount to order table ($\text{PricePerUnit} * \text{Qty}$)
- Business Value: Simplifies downstream aggregation and analytics
- Priority: Medium

BR-ETL-006 — Data Quality: Null and Duplicate Removal

- Description: Remove nulls and duplicates from customer and order tables
- Business Value: Improves analytical accuracy and reliability
- Priority: High

BR-ETL-007 — Create Ordersummary Table

- Description: Create ordersummary table in catalog=gen_ai_poc_databrickscoe, schema=sdlc_wizard if not exists
- Business Value: Centralized storage for enriched order data
- Priority: High

BR-ETL-008 — SCD Type 2 on Ordersummary

- Description: Apply SCD Type 2 methodology to ordersummary based on updates in customer table
- Business Value: Ensures historical accuracy and traceability of data changes
- Priority: High

BR-ETL-009 — Create CustomerAggregateSpend Table

- Description: Create customeraggregatespend table with columns Name, TotalAmount, Date if not exists
- Business Value: Enables spend analysis and reporting
- Priority: Medium

BR-ETL-010 — Aggregate Customer Spend

- Description: Aggregate TotalAmount grouped by Name and Date from ordersummary and load into customeraggregatespend
- Business Value: Facilitates spend tracking for business insights
- Priority: Medium

BR-ETL-011 — Delta Live Tables Implementation

- Description: Implement the entire processing steps using Delta Live Tables
- Business Value: Simplifies workflow orchestration and monitoring
- Priority: High

Assumptions & Constraints

- Source data is available and accessible via specified CSV file paths
- Only defined table schemas will be enforced
- Delta Lake infrastructure is operational
- Predefined storage locations are available

Success Metrics

- Automated, successful processing from CSV to target tables
- No null or duplicate records in output tables
- Accurate aggregation in customeraggregatespend table

- Auditable SCD Type 2 history maintained

Risks & Mitigations

- Source data format inconsistencies — Mitigation: Strict schema enforcement and validation
- Poor data quality from source — Mitigation: Elimination of nulls/duplicates in preprocessing
- Infrastructure or access issues — Mitigation: Pre-execution checks and ongoing monitoring

Approval

- Business Owner: Sarah Marshall
- Data Engineering Lead: Ravi Patel
- Compliance Review: Janet Lewis
- Date: 2024-06-15

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
customerdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Customer master source
orderdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Order transaction source

Appendix B: Target Definitions

	Columns
databrickscoe.sdlc_wizard	OrderId, ItemName, PricePerUnit, Qty, Date, CustId, Name, EmailId, Region, TotalAmount, scd_valid_from, scd_valid_to
databrickscoe.sdlc_wizard	Name, TotalAmount, Date

Appendix C: Transformations

Transformation	Input	Output	Logic
Enforce customer schema	customerdata	delta customer table	Map and validate CustId, Name, EmailId, Region
Enforce order schema	orderdata	delta order table	Map and validate OrderId, ItemName, PricePerUnit, Qty, Date
Calculate TotalAmount	order table	order table	PricePerUnit * Qty
Remove nulls/duplicates	both tables	cleaned tables	Drop null and duplicate rows
Join & load ordersummary	cleaned tables	ordersummary	Join on CustId, enrich order with customer data, add fields
SCD Type 2 implementation	ordersummary	ordersummary	Track historical changes using SCD columns
Aggregate spend	ordersummary	customeraggregatespend	Group by Name, Date and sum TotalAmount

Appendix D: Business Rules

Rule ID	Area	Description	Source
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BR-ETL-003	Data Quality	Enforce schema for customer table	BRD
BR-ETL-004	Data Quality	Enforce schema for order table	BRD
BR-ETL-006	Data Quality	Remove all null values and duplicates	BRD
BR-ETL-008	SCD	Maintain SCD Type 2 history on ordersummary	BRD

Appendix E: Process Flow

Step	Description	Dependency
1	Load source customer and order data	Source availability
2	Schema enforcement and data cleaning	Step 1
3	Add TotalAmount, remove nulls/duplicates	Step 2
4	Load and join to create ordersummary	Step 3
5	Apply SCD Type 2 logic to ordersummary	Step 4
6	Aggregate spend and load final table	Step 5



